Определённые интегралы

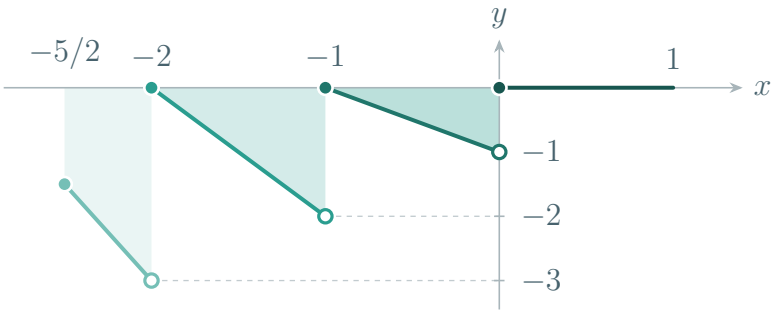
Разбор



Вариант 1

1. $\displaystyle\int_{-5/2}^{1} [x]\{x\}\,dx.$

Ответ: $-\dfrac{21}{8}$



$$\int_{-5/2}^{1} [x]\{x\}\,dx = \int_{-5/2}^{-2} (-3)(x+3)\,dx + \int_{-2}^{-1} (-2)(x+2)\,dx + \int_{-1}^{0} (-1)(x+1)\,dx + \int_{0}^{1} 0\,dx = -\frac{9}{8} - 1 - \frac{1}{2} + 0 = -\frac{21}{8}.$$

2. $\displaystyle\int_{0}^{1} \frac{239x^2+1}{x^3+1}\,dx.$

Ответ: $80\ln 2 + \dfrac{\pi}{3\sqrt{3}}$

Раскладываем дробь на простейшие **методом неопределённых коэффициентов**:

$$\frac{239x^2+1}{x^3+1} = \frac{239x^2+1}{(x+1)(x^2-x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2-x+1},$$

$$239x^2+1 = A(x^2-x+1) + (Bx+C)(x+1) = (A+B)x^2 + (B+C-A)x + (A+C),$$

$$\begin{cases} A+B=239 \\ B+C-A=0 \\ A+C=1 \end{cases} \quad \Longrightarrow \quad A=80, \quad B=159, \quad C=-79.$$

Выделяем в числителе производную знаменателя $2x-1$:

$$159x-79=\frac{159}{2}(2x-1)+\frac{1}{2},$$

$$\int_0^1 \frac{239x^2+1}{x^3+1}\,dx=\int_0^1 \frac{80\,dx}{x+1}+\frac{159}{2}\int_0^1 \frac{(2x-1)\,dx}{x^2-x+1}$$
$$+\frac{1}{2}\int_0^1 \frac{dx}{(x-\frac{1}{2})^2+\frac{3}{4}}=80\ln 2+\frac{159}{2}\ln(x^2-x+1)\Big|_0^1$$
$$+\frac{1}{\sqrt{3}}\operatorname{arctg}\frac{2x-1}{\sqrt{3}}\Big|_0^1=80\ln 2+0+\frac{\pi}{3\sqrt{3}}.$$

3. $\displaystyle\int_{\pi/4}^{5\pi/12}\left(\sqrt[4]{\operatorname{tg} x}+\sqrt[4]{\operatorname{ctg} x}\right)^2 dx.$

Ответ: $\dfrac{\pi\sqrt{2}}{4}+\dfrac{\pi}{3}$

$$\left(\sqrt[4]{\operatorname{tg} x}+\sqrt[4]{\operatorname{ctg} x}\right)^2=\sqrt{\operatorname{tg} x}+\sqrt{\operatorname{ctg} x}+2,$$

$$\int_{\pi/4}^{5\pi/12}\left(\sqrt{\operatorname{tg} x}+\sqrt{\operatorname{ctg} x}\right)dx=\int_{\pi/4}^{5\pi/12}\frac{(\sin x+\cos x)\,dx}{\sqrt{\sin x\cos x}}$$
$$=\int_{\pi/4}^{5\pi/12}\frac{d(\sin x-\cos x)}{\sqrt{\sin x\cos x}}=\int_{\pi/4}^{5\pi/12}\frac{\sqrt{2}\,d(\sin x-\cos x)}{\sqrt{1-(\sin x-\cos x)^2}},$$

$$u=\sin x-\cos x,$$

$$\int_{\pi/4}^{5\pi/12}\left(\sqrt{\operatorname{tg} x}+\sqrt{\operatorname{ctg} x}\right)dx=\int_0^{\sqrt{2}/2}\frac{\sqrt{2}\,du}{\sqrt{1-u^2}}$$
$$=\sqrt{2}\arcsin u\Big|_0^{\sqrt{2}/2}=\sqrt{2}\arcsin\frac{\sqrt{2}}{2}=\frac{\pi\sqrt{2}}{4},$$

$$\int_{\pi/4}^{5\pi/12} \left(\sqrt[4]{\operatorname{tg} x} + \sqrt[4]{\operatorname{ctg} x}\right)^2 dx = \frac{\pi\sqrt{2}}{4} + 2 \cdot \frac{\pi}{6} = \frac{\pi\sqrt{2}}{4} + \frac{\pi}{3}.$$

4. $\displaystyle\int_0^{\pi/2} \cos^8 x \, \cos 8x \, dx.$

Ответ: $\dfrac{\pi}{512}$

$$\int_0^{\pi/2} \cos^8 x \cos 8x \, dx = \int_0^{\pi/2} \cos^8 x \, d\Big(\frac{\sin 8x}{8}\Big)$$

$$= \int_0^{\pi/2} \cos^7 x \, \sin x \sin 8x \, dx.$$

Произведение $\sin x \sin 8x$ выражаем из косинуса разности:

$$\sin x \sin 8x = \cos(8x - x) - \cos 8x \cos x = \cos 7x - \cos x \cos 8x,$$

$$\int_0^{\pi/2} \cos^8 x \cos 8x \, dx = \int_0^{\pi/2} \cos^7 x \, (\cos 7x - \cos x \cos 8x) \, dx,$$

$$\int_0^{\pi/2} \cos^8 x \cos 8x \, dx = \frac{1}{2} \int_0^{\pi/2} \cos^7 x \cos 7x \, dx = \ldots$$

$$= \frac{1}{2^8} \int_0^{\pi/2} dx = \frac{\pi}{512}.$$

5. $\displaystyle\int_0^1 \frac{x^3 - 1}{\ln x} \, dx.$

Ответ: $\ln 4$

Применяем **трюк Фейнмана**: заменяем число 3 на a.

$$F(a) = \int_0^1 \frac{x^a - 1}{\ln x} \, dx, \qquad F(0) = 0,$$

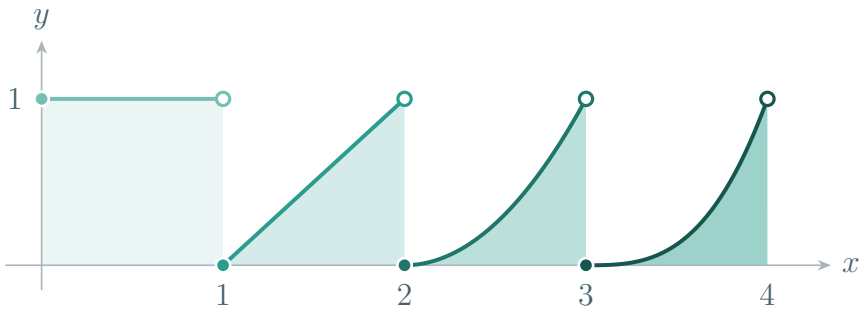
$$F'(a) = \int_0^1 \frac{x^a \ln x}{\ln x} \, dx = \int_0^1 x^a \, dx = \frac{1}{a+1},$$

$$F(a) = \ln(a+1), \qquad F(3) = \ln 4.$$

Вариант 2

1. $\displaystyle\int_0^4 \{x\}^{[x]}\,dx.$

Ответ: $\dfrac{25}{12}$



$$\int_0^4 \{x\}^{[x]}\,dx = \int_0^1 1\,dx + \int_1^2 (x-1)\,dx + \int_2^3 (x-2)^2\,dx + \int_3^4 (x-3)^3\,dx = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12}.$$

2. $\displaystyle\int_0^1 \frac{239x^2-1}{x^3+x^2+x+1}\,dx.$

Ответ: $179\ln 2 - 30\pi$

Раскладываем дробь на простейшие **методом неопределённых коэффициентов**:

$$\frac{239x^2-1}{x^3+x^2+x+1} = \frac{239x^2-1}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1},$$

$$239x^2-1 = A(x^2+1) + (Bx+C)(x+1) = (A+B)x^2 + (B+C)x + (A+C),$$

$$\begin{cases} A+B=239 \\ B+C=0 \\ A+C=-1 \end{cases} \quad\Longrightarrow\quad A=119,\quad B=120,\quad C=-120,$$

$$\int_0^1 \frac{239x^2-1}{x^3+x^2+x+1}\,dx = \int_0^1 \Big(\frac{119}{x+1}+\frac{120x}{x^2+1}-\frac{120}{x^2+1}\Big)dx$$

$$= 119\ln 2 + 60\ln(x^2+1)\Big|_0^1 - 120\operatorname{arctg} x\Big|_0^1$$

$$= 119\ln 2 + 60\ln 2 - 30\pi.$$

3. $\displaystyle\int_{\pi/12}^{\pi/4} \big(\sqrt{\operatorname{tg} x}+\sqrt{\operatorname{ctg} x}\big)\,dx.$

Ответ: $\dfrac{\pi\sqrt{2}}{4}$

$$\int_{\pi/12}^{\pi/4} \big(\sqrt{\operatorname{tg} x}+\sqrt{\operatorname{ctg} x}\big)\,dx = \int_{\pi/12}^{\pi/4} \frac{(\sin x+\cos x)\,dx}{\sqrt{\sin x\cos x}}$$

$$= \int_{\pi/12}^{\pi/4} \frac{d(\sin x-\cos x)}{\sqrt{\sin x\cos x}} = \int_{\pi/12}^{\pi/4} \frac{\sqrt{2}\,d(\sin x-\cos x)}{\sqrt{1-(\sin x-\cos x)^2}},$$

$$u=\sin x-\cos x,$$

$$\int_{\pi/12}^{\pi/4} \big(\sqrt{\operatorname{tg} x}+\sqrt{\operatorname{ctg} x}\big)\,dx = \int_{-\sqrt{2}/2}^{0} \frac{\sqrt{2}\,du}{\sqrt{1-u^2}}$$

$$= \sqrt{2}\arcsin u\,\Big|_{-\sqrt{2}/2}^{0} = -\sqrt{2}\arcsin\Big(-\frac{\sqrt{2}}{2}\Big) = \frac{\pi\sqrt{2}}{4}.$$

4. $\displaystyle\int_0^{\pi/2} \sin^{10} x\, \cos 10x\, dx.$

Ответ: $\displaystyle -\frac{\pi}{2048}$

$$\int_0^{\pi/2} \sin^{10} x \cos 10x\, dx = \int_0^{\pi/2} \sin^{10} x\, d\Big(\frac{\sin 10x}{10}\Big)$$

$$= -\int_0^{\pi/2} \sin^{9} x\, \cos x \sin 10x\, dx.$$

Произведение $\cos x \sin 10x$ выражаем из синуса разности:

$$\cos x \sin 10x = \sin(10x - x) + \sin x \cos 10x = \sin 9x + \sin x \cos 10x,$$

$$\int_0^{\pi/2} \sin^{10} x \cos 10x\, dx = -\int_0^{\pi/2} \sin^{9} x\, (\sin 9x + \sin x \cos 10x)\, dx,$$

$$\int_0^{\pi/2} \sin^{10} x \cos 10x\, dx = -\frac{1}{2}\int_0^{\pi/2} \sin^{9} x \sin 9x\, dx,$$

$$\int_0^{\pi/2} \sin^{9} x \sin 9x\, dx = \int_0^{\pi/2} \sin^{9} x\, d\Big(-\frac{\cos 9x}{9}\Big)$$

$$= \int_0^{\pi/2} \sin^{8} x\, \cos x \cos 9x\, dx.$$

Произведение $\cos x \cos 9x$ выражаем из косинуса разности:

$$\cos x \cos 9x = \cos(9x - x) - \sin x \sin 9x = \cos 8x - \sin x \sin 9x,$$

$$\int_0^{\pi/2} \sin^{9} x \sin 9x\, dx = \int_0^{\pi/2} \sin^{8} x\, (\cos 8x - \sin x \sin 9x)\, dx,$$

$$\int_0^{\pi/2} \sin^{9} x \sin 9x\, dx = \frac{1}{2}\int_0^{\pi/2} \sin^{8} x \cos 8x\, dx,$$

$$\int_0^{\pi/2} \sin^{10} x \cos 10x\, dx = -\frac{1}{4}\int_0^{\pi/2} \sin^{8} x \cos 8x\, dx = \ldots$$

$$= \Big(-\frac{1}{4}\Big)^5 \int_0^{\pi/2} dx = -\frac{\pi}{2048}.$$

5. $\displaystyle\int_0^1 \frac{x^5 - x^2}{\ln x}\,dx.$

Ответ: $\ln 2$

Применяем **трюк Фейнмана**: заменяем число 5 на a.

$$F(a) = \int_0^1 \frac{x^a - x^2}{\ln x}\,dx, \qquad F(2) = 0,$$

$$F'(a) = \int_0^1 \frac{x^a \ln x}{\ln x}\,dx = \int_0^1 x^a\,dx = \frac{1}{a+1},$$

$$F(a) = \ln\frac{a+1}{3}, \qquad F(5) = \ln 2.$$